

```
In [1]: l1=[2,3,34,4,4,5,6,66,7]
```

```
In [2]: l1
```

```
Out[2]: [2, 3, 34, 4, 4, 5, 6, 66, 7]
```

```
In [3]: l1
```

```
Out[3]: [2, 3, 34, 4, 4, 5, 6, 66, 7]
```

```
In [4]: l1[1:10:2]
```

```
Out[4]: [3, 4, 5, 66]
```

```
In [6]: all(l1)
```

```
Out[6]: True
```

```
In [8]: # python is a programming language has syntax and only practise no theory  
#ml - theory + practical  
#stat - theory + practical
```

TUPLE

```
In [9]: t = ()
```

```
In [10]: t
```

```
Out[10]: ()
```

```
In [11]: print(type(t))
```

```
<class 'tuple'>
```

```
In [12]: t1 = (10,20,30,40,40)
```

```
In [13]: t1
```

```
Out[13]: (10, 20, 30, 40, 40)
```

```
In [14]: len(t1)
```

```
Out[14]: 5
```

```
In [16]: t1.count(10) # list is mutable and ashable
```

```
Out[16]: 1
```

```
In [17]: t1.count(40)
```

Out[17]: 2

In [18]: `t1.index(10)`

Out[18]: 0

In [22]: `l5=['a','b','c',10]`

In [23]: `l5[1] = 10`

In [24]: `l5`

Out[24]: ['a', 10, 'c', 10]

In [25]: `t2=(100,True,(1,2,3),2.5,'jai ',(10,20,50))`

In [26]: `t2`

Out[26]: (100, True, (1, 2, 3), 2.5, 'jai ', (10, 20, 50))

In [27]: `print(t)
print(t1)
print(t2)`

()
(10, 20, 30, 40, 40)
(100, True, (1, 2, 3), 2.5, 'jai ', (10, 20, 50))

In [28]: `t1`

Out[28]: (10, 20, 30, 40, 40)

In [30]: `t1[1]= 100` *# tuple is immutable or ashable duplcte allowed multiple data types i*

```
-----
TypeError                                Traceback (most recent call last)
Cell In[30], line 1
----> 1 t1[1]= 100

TypeError: 'tuple' object does not support item assignment
```

In [31]: `t2`

Out[31]: (100, True, (1, 2, 3), 2.5, 'jai ', (10, 20, 50))

tuple is immutable or ashable

- duplcte allowed
- multiple data types
- indexing and slicing allowed

- only two functioncount index

```
In [32]: icic =(4348820,"iqmpk8912",32000,98765)
```

```
In [38]: t1
```

```
Out[38]: (10, 20, 30, 40, 40)
```

```
In [39]: t4=t1*3 # as tuple immutable it repeats 3 times
```

```
In [40]: t4
```

```
Out[40]: (10, 20, 30, 40, 40, 10, 20, 30, 40, 40, 10, 20, 30, 40, 40)
```

```
In [41]: t4
```

```
Out[41]: (10, 20, 30, 40, 40, 10, 20, 30, 40, 40, 10, 20, 30, 40, 40)
```

```
In [42]: t4[:]
```

```
Out[42]: (10, 20, 30, 40, 40, 10, 20, 30, 40, 40, 10, 20, 30, 40, 40)
```

```
In [43]: t4[2:8]
```

```
Out[43]: (30, 40, 40, 10, 20, 30)
```

```
In [44]: t4[::2]
```

```
Out[44]: (10, 30, 40, 20, 40, 10, 30, 40)
```

```
In [45]: t4[0:10:2]
```

```
Out[45]: (10, 30, 40, 20, 40)
```

```
In [46]: t4
```

```
Out[46]: (10, 20, 30, 40, 40, 10, 20, 30, 40, 40, 10, 20, 30, 40, 40)
```

```
In [47]: t4[2:]
```

```
Out[47]: (30, 40, 40, 10, 20, 30, 40, 40, 10, 20, 30, 40, 40)
```

```
In [48]: t4.remove(20) # onlt two functions index and count
```

```
-----
AttributeError                                Traceback (most recent call last)
Cell In[48], line 1
----> 1 t4.remove(20)

AttributeError: 'tuple' object has no attribute 'remove'
```

```
In [54]: x =5  
y =2  
print(x//y)
```

2

set

distrubution concept born from probability

- probability used in stock market
- flight booking irctc

```
In [55]: s={}
```

```
In [56]: s
```

```
Out[56]: {}
```

```
In [58]: type(s)
```

```
Out[58]: dict
```

```
In [ ]:
```